

Module 3

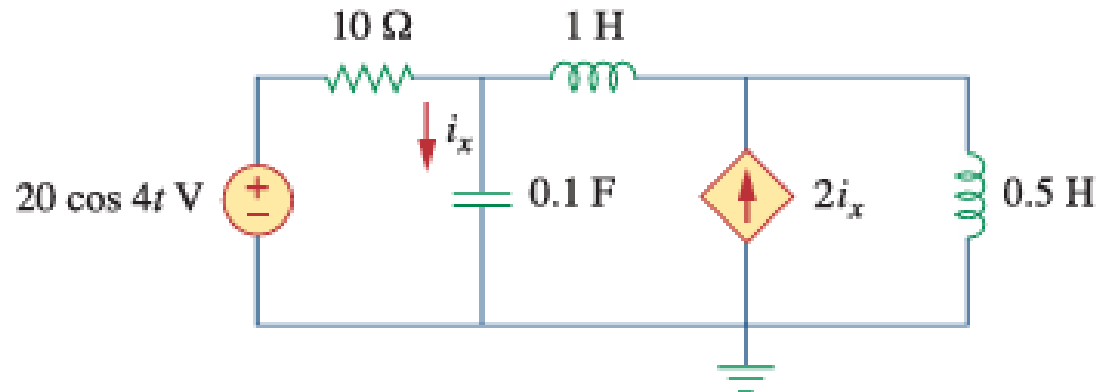
Sinusoidal Steady-State Analysis

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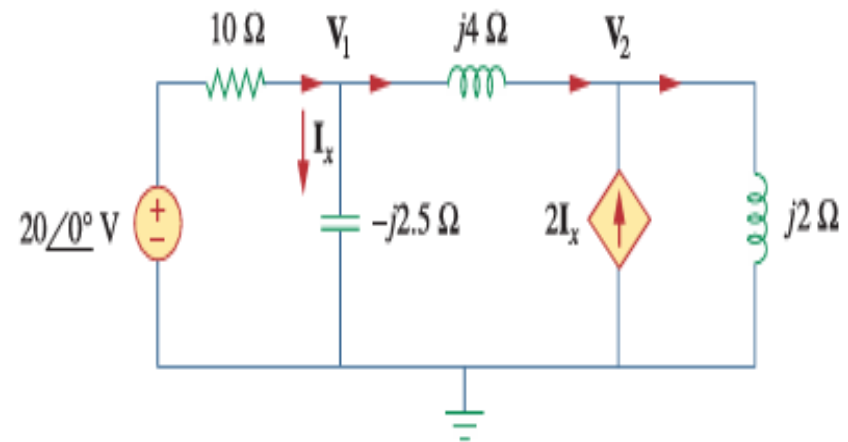
Nodal analysis

1. Find I_x in the circuit using nodal analysis method.



We first convert the circuit to the frequency domain:

$$\begin{aligned} 20 \cos 4t &\Rightarrow 20 \angle 0^\circ, & \omega &= 4 \text{ rad/s} \\ 1 \text{ H} &\Rightarrow j\omega L = j4 \\ 0.5 \text{ H} &\Rightarrow j\omega L = j2 \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2.5 \end{aligned}$$



Nodal analysis

Applying KCL at node 1,

$$\frac{20 - \mathbf{V}_1}{10} = \frac{\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4}$$

or

$$(1 + j1.5)\mathbf{V}_1 + j2.5\mathbf{V}_2 = 20$$

At node 2,

$$2\mathbf{I}_x + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

But $\mathbf{I}_x = \mathbf{V}_1 / -j2.5$. Substituting this gives

$$\frac{2\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

By simplifying, we get

$$11\mathbf{V}_1 + 15\mathbf{V}_2 = 0$$

$$\mathbf{V}_1 = \frac{\Delta_1}{\Delta} = \frac{300}{15 - j5} = 18.97 \angle 18.43^\circ \text{ V}$$

$$\mathbf{V}_2 = \frac{\Delta_2}{\Delta} = \frac{-220}{15 - j5} = 13.91 \angle 198.3^\circ \text{ V}$$

The current $\mathbf{I}_x$ is given by

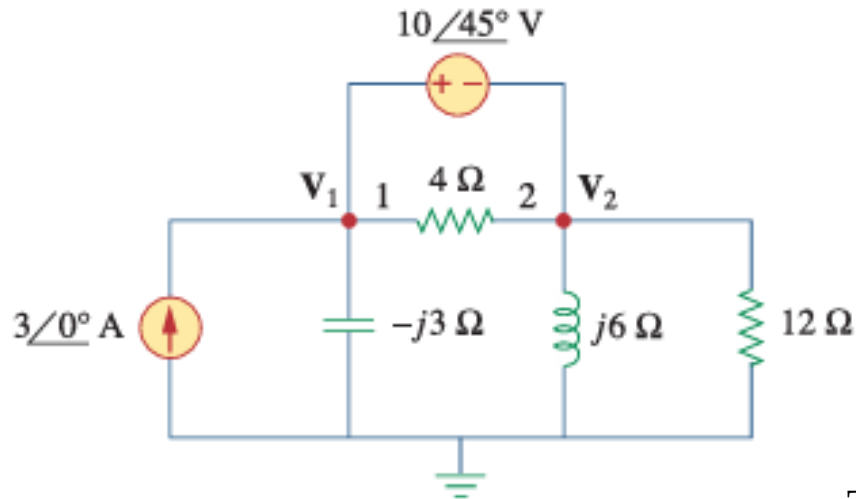
$$\mathbf{I}_x = \frac{\mathbf{V}_1}{-j2.5} = \frac{18.97 \angle 18.43^\circ}{2.5 \angle -90^\circ} = 7.59 \angle 108.4^\circ \text{ A}$$

Transforming this to the time domain,

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

Nodal analysis

2. Compute V_1 and V_2 in the circuit using nodal analysis method.



Applying KCL at supernode gives,

$$3 = \frac{\mathbf{V}_1}{-j3} + \frac{\mathbf{V}_2}{j6} + \frac{\mathbf{V}_2}{12}$$

$$36 = j4\mathbf{V}_1 + (1 - j2)\mathbf{V}_2 \quad \text{---1}$$

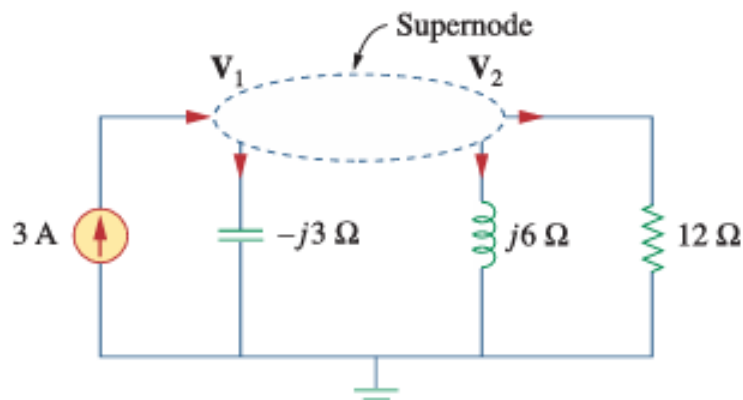
The voltage source is between nodes V_1 and V_2

$$\mathbf{V}_1 = \mathbf{V}_2 + 10\angle 45^\circ \quad \text{---2}$$

Substituting eqn.2 in eqn.1 gives,

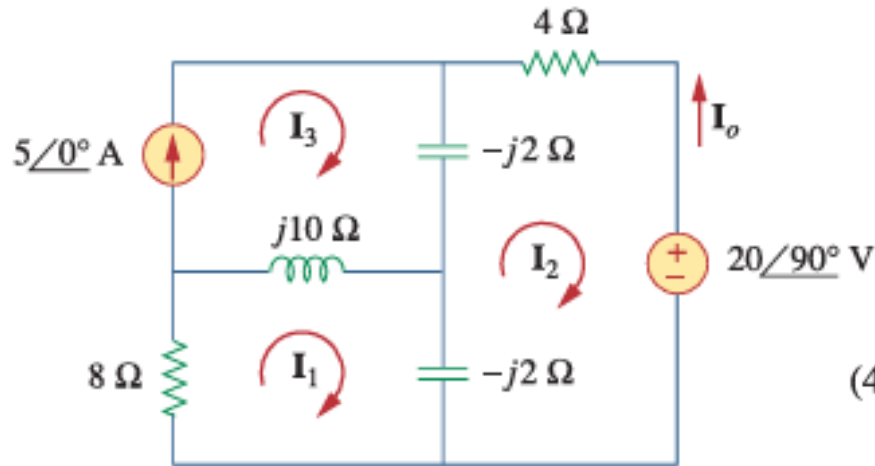
$$36 - 40\angle 135^\circ = (1 + j2)\mathbf{V}_2 \Rightarrow \mathbf{V}_2 = 31.41\angle -87.18^\circ \text{ V}$$

$$\mathbf{V}_1 = \mathbf{V}_2 + 10\angle 45^\circ = 25.78\angle -70.48^\circ \text{ V}$$



Mesh analysis

1. Compute the current I_o in the circuit using mesh analysis method.



Applying KVL to all the meshes gives,

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0 \quad \text{---1}$$

$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0 \quad \text{---2}$$

From mesh 3, I_3 is 5 A, hence, eqn.1 and eqn.2 becomes,.

$$(8 + j8)\mathbf{I}_1 + j2\mathbf{I}_2 = j50$$

$$j2\mathbf{I}_1 + (4 - j4)\mathbf{I}_2 = -j20 - j10$$

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{416.17\angle -35.22^\circ}{68} = 6.12\angle -35.22^\circ \text{ A}$$

$$\mathbf{I}_o = -\mathbf{I}_2 = 6.12\angle 144.78^\circ \text{ A}$$

Next class: SuperpositionTheorem